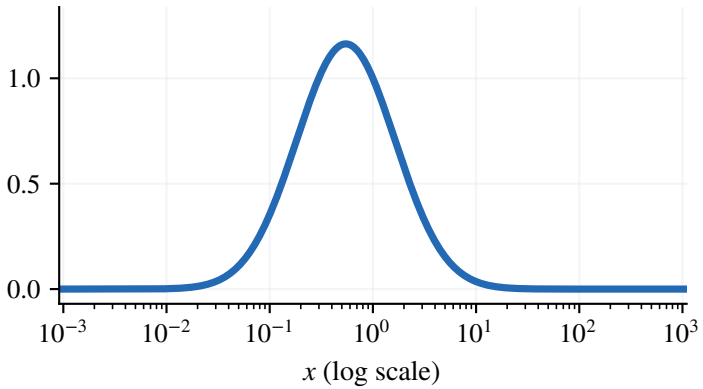
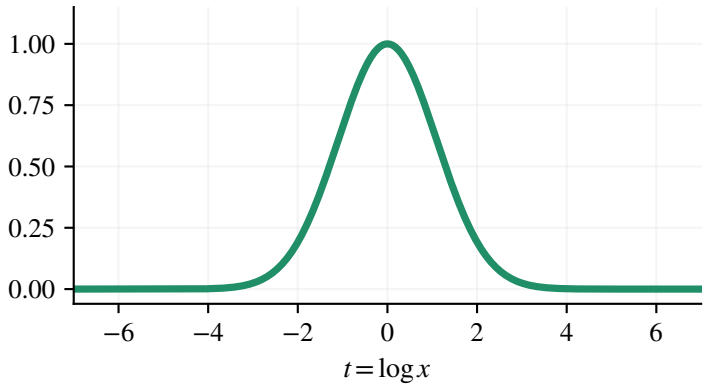
MELLIN / FOURIER PIPELINE

$$f(x) \to g(t) \to \hat{g}(\lambda) \to a(\lambda)\hat{g}(\lambda) \to h(x)$$

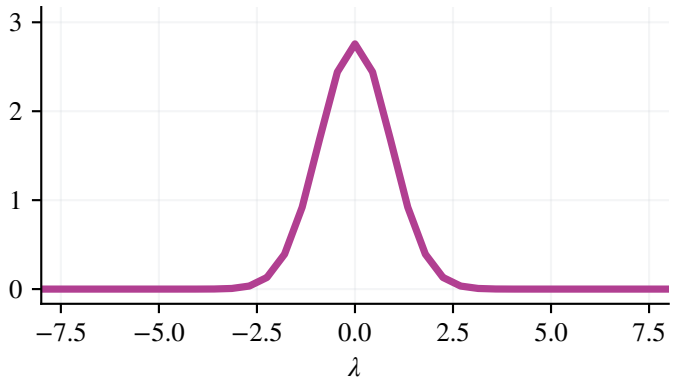
1 INPUT $f(x)$



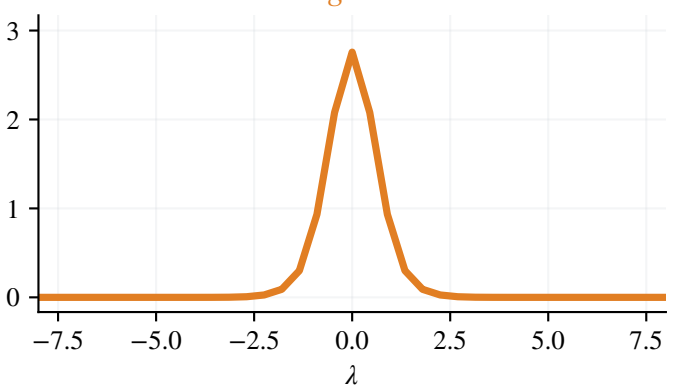
2 LOG FUNCTION $g(t)$



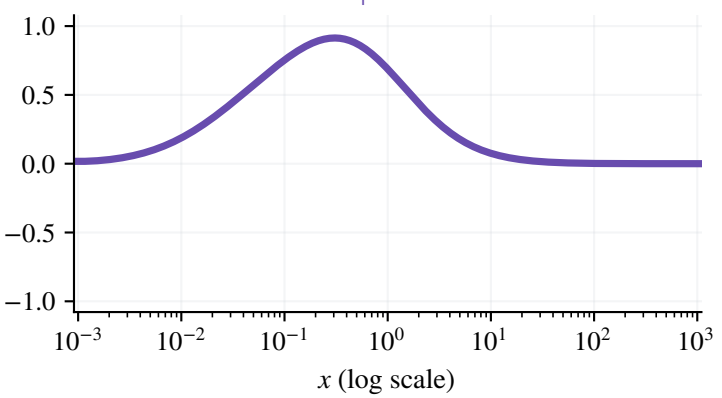
3 FOURIER $\hat{g}$



4 MULTIPLIED $a\hat{g}$



5 OUTPUT ON R_+



Numerical demo: $a(\lambda) = 0.20 + 0.80e^{-(\lambda/0.70)^2/2}$. Paper identity: $a_{\mathrm{K25}}(\lambda) = \coth \pi(\lambda + i\nu)$.

$E_\nu f(t) = e^{\nu t} f(e^t)$; $M^0(a) = E_\nu^{-1} \mathcal{F}^{-1} a \mathcal{F} E_\nu$. Numerical illustration, not a proof.